A Pointwise Digit Certificate and a Cyclic-Modulus Counterexample

Qichao Wang
Hebei University of Technology, Tianjin, China
Daoyu Dong
University of Electronic Science and Technology of China, Chengdu, China

November 11, 2026

Abstract

An explicit whole-block rotation is a square in every local field associated with the full cyclic modulus and the decimal base, but is not an integer square. This disproves the sufficiency of that family of congruence tests at every precision. A real-interval test with the complete roots of the actual decimal suffix excludes the candidate; one added digit from a middle block eliminates four surviving square completions. This certificate does not settle Kashihara's conjecture.

Write $W_j = 1\|\cdots\|j$, $D(j) = \mathrm{digits}(W_j)$, and $W_0 = D(0) = 0$. Moving the first k whole integer blocks to the end gives

$$R_{m,k} = 10^{D(k)}W_m - (10^{D(m)} - 1)W_k, \qquad 0 \le k < m. \tag{1}$$

The previous bounds $O(X)$ for square pairs and $O(X^2/\log X)$ for perfect-power pairs permit isolated exceptions. No counting bound is used below. All integer positions in (1), including the middle blocks, retain their prescribed values.

1 A complete cyclic modulus does not suffice

Proposition 1.1. *The actual rotation*

$$R = R_{16,1} = 23456789101112131415161, \qquad \Delta = 10^{23} - 1,$$

is a square in $\mathbb{Q}_p$ for every prime $p \mid 10\Delta$, but is not an integer square. Consequently it is a square modulo $10^a\Delta^b$ for every pair of nonnegative integers a, b. Thus even the complete cyclic modulus together with arbitrarily precise decimal congruences cannot reject every individual rotation.

Proof. The word is $2\|3\|\cdots\|16\|1$, with 23 digits, and (1) reads $R = 10W_{16} - \Delta$. Direct integer arithmetic gives

$$r = 24515341518503348803133, \qquad r^2 \equiv R \pmod{\Delta}, \qquad \gcd(2r, \Delta) = 1.$$

For every prime divisor p of Δ, this is a nonsingular root modulo p, which lifts to all p-adic precisions by Hensel's lemma. No factorization of Δ is needed. Also $R \equiv 1 \pmod 8$ and $R \equiv 1 \pmod 5$, giving square roots in $\mathbb{Q}_2$ and $\mathbb{Q}_5$. These are integral local roots. The Chinese remainder theorem proves the assertion for $10^a\Delta^b$. Section 3 proves that R is not an integer square using an actual middle digit. □

This is a counterexample to the sufficiency of these local conditions, not to Kashihara's conjecture. In particular, retaining all middle digits in a congruence is different from forcing a root of the required real size. No claim that every pointwise method fails follows.

2 Exact compatibility with the real interval

Let a positive integer R have E digits. Prescribe its first h digits P and its last q digits $U = R \bmod 10^q$, with $q \ge 3$. The digits of U are taken from the actual word, and may extend past the original moved prefix into its middle blocks. Put

$$A = \left\lceil \sqrt{P10^{E-h}} \right\rceil, \qquad B = \left\lfloor \sqrt{(P+1)10^{E-h} - 1} \right\rfloor.$$

Assume $v_2(U), v_5(U) < q$. An odd valuation excludes squares immediately. Otherwise write $v_2(U) = 2a$, $v_5(U) = 2b$, and

$$z = 2^a5^b, \qquad H = 10^q/z.$$

Solving the two unit congruences at 2 and 5 and applying CRT gives a complete set $\mathcal{A} \subset [0, H)$ of at most eight root classes:

$$y^2 \equiv U \pmod{10^q} \iff y \equiv r \pmod{H} \text{ for some } r \in \mathcal{A}. \tag{2}$$

Indeed, division by p^{2a} or p^{2b} gives unit square roots, at most four at 2 and two at 5. Multiplying back gives root moduli 2^{q-a} and 5^{q-b}, whose product is H.

Lemma 2.1 (Pointwise interval test)**.** *A square with the prescribed length, prefix and suffix exists if and only if, for some* $r \in \mathcal{A}$*,*

$$\max\left(0, \left\lceil \frac{A-r}{H} \right\rceil\right) \;\le\; \left\lfloor \frac{B-r}{H} \right\rfloor.$$

If $B - A < H$*, at most eight possible positive roots occur. Failure of all these inequalities excludes the particular rotation supplying the digits.*

Proof. The prefix and length are equivalent to $A \le y \le B$; the suffix is equivalent to (2). Write $y = r + jH$, with $j \ge 0$, and solve the two interval inequalities for j. An interval shorter than H contains at most one integer from each root class. □

The test is exact for the selected windows. If it leaves roots, the additional actual middle digits must be checked; their values cannot be filled freely. Increasing q preserves every genuine square candidate. At $q = E$, the test is an exact square decision, since $y^2 < 10^E$ and $y^2 \equiv R \pmod{10^E}$ force equality. This termination observation does not prove uniform exclusion for the infinite rotation family.

3 One middle digit excludes the isolated candidate

For Proposition 1.1, the first digit is 2, so

$$[A, B] = [141421356238,\ 173205080756].$$

At precision $q = 12$, the actual suffix is

$$U = 112131415161 = 1\|12\|13\|14\|15\|16\|1.$$

Its first displayed 1 is the final digit of the middle block 11; it is not part of the original moved prefix $W_1 = 1$. Here $H = 10^{12}$. The complete eight roots modulo H are

55164164619	104480570869	395519429131	444835835381
555164164619	604480570869	895519429131	944835835381.

Each entry squares to U modulo 10^{12}. They are distinct, and the unit-root bound of eight proves completeness. None belongs to $[A, B]$; also $B < H$, so no positive translate of a listed root can belong to the interval. Lemma 2.1 proves that R is not a square.

The extra middle digit is substantive. The weaker 11-digit suffix 12131415161 and the same first digit and total length admit the following four squares:

Root	Square	Twelfth digit from the right
144835835381	20977419210512131415161	5
145519429131	21175904254612131415161	6
154480570869	23864246776012131415161	0
155164164619	24075917981912131415161	9

All four have 23 digits, start with 2, and share the actual last 11 digits. The next actual digit is 1, whereas these completions have $5, 6, 0, 9$. The complete 12-digit calculation above excludes every completion, not merely these four examples.

4 Verification and the remaining problem

The independent Python verifier checks every displayed certificate exactly and compares the GMP/OpenMP implementation with the earlier Python root solver and small exhaustive intervals. The window and Hensel/CRT tools are standard; no priority claim is made. No uniform theorem proves that the interval test rejects every rotation, and square tests do not exclude odd prime powers. Kashihara's conjecture remains open; the existing sparsity bounds have not been strengthened.